

Notation

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Table 1: Notation used in this book

symbol	meaning	LaTeX
$\neg$	not	<code>\neg</code>
$\forall$	all	<code>\forall</code>
$\exists$	some	<code>\exists</code>
$\cup$	union, “or”	<code>\cup</code>
$\cap$	intersection, “and”	<code>\cap</code>
$ $	given, conditional on	<code>\mid</code> , <code> </code>
$\sum$	sum	<code>\sum</code>
$\prod$	product	<code>\prod</code>
μ	mean	<code>\mu</code>
E	expectation ¹	<code>\mathbb{E}</code>
$x^\top$	transpose of x	<code>x^{\top}</code>
$'$	transpose or derivative ²	<code>'</code>
$\perp$	independent ³	

¹[probability.qmd#def-expectation](#)

²depending on whether it is applied to a matrix or a function

³[probability.qmd#def-indpt](#)

symbol	meaning	LaTeX
$\therefore$	therefore, thus	<code>\therefore</code>
η	linear component of a GLM ⁴	<code>\eta</code>
$\lfloor x \rfloor$	floor of x : largest integer smaller than x	<code>\lfloor x \rfloor</code>
$\lceil x \rceil$	ceiling of x : smallest integer larger than x	<code>\lceil x \rceil</code>
$\mathbb{1}_A(x), \mathbb{1}(P)$	indicator function (Section 4): 1 if condition holds, 0 otherwise	<code>\indic{A}(x), \indic{P}</code>

1 Information matrices

There is no consistency in the notation for observed and expected information matrices (see Table 2).

Table 2: notation for information matrices

book	observed information	expected information
Dobson and Barnett (2018)	U'	$\mathfrak{J}$
Dunn and Smyth (2018)	$\mathfrak{J}$	$\mathcal{J}$
McLachlan and Krishnan (2007)	I	$\mathcal{J}$
Wood (2017)	$\hat{I}$	$\mathcal{J}$

These notes currently have a mixture of notations, depending on my whims and what reference I had last looked at. Eventually, I will try to standardize my notation to I for observed information and $\mathcal{J}$ for expected information.

2 Percent sign (“%”)

The percent sign “%” is just a shorthand for “/100”. The word “percent” comes from the Latin “per centum”; “centum” is Latin for 100, so “percent” means “per hundred” (c.f., <https://en.wikipedia.org/wiki/Percentage>)

So, contrary to what you may have learned previously, $10\% = 0.1$ is a true and correct equality, just as $10\text{kg} = 10,000\text{g}$ is true and correct.

i Proof

Proof.

$$\begin{aligned}
 10\% &= 10/100 \\
 &= \frac{10}{100} \\
 &= 0.1
 \end{aligned}$$

□

You are welcome to switch between decimal and percent notation freely; just make sure you execute it correctly.

3 Proofs

We can use any of:

⁴https://en.wikipedia.org/wiki/Generalized_linear_model#:~:text=The%20linear%20predictor%20is%20the,data%20through%20the%20link%20function

- $\therefore$ (`\therefore` in LaTeX),
- $\Rightarrow$ (`\Rightarrow` in LaTeX),
- $\models$ (`\models` in LaTeX)

to denote logical entailments (deductive consequences).

Let's save $\rightarrow$ (`\rightarrow` in LaTeX) for convergence results.

See Proof Writing⁵ for general guidance on how to present proofs and derivations.

4 Indicator functions

An **indicator function** is a mathematical function that signals whether an element belongs to a specified set, or whether a given logical condition is satisfied. In statistics and epidemiology, indicator functions are ubiquitous: they represent binary variables, censor and event indicators in survival analysis, membership in subpopulations, and domain restrictions in integrals and sums.

Despite their conceptual simplicity, notation for indicator functions varies substantially across textbooks, research papers, and subfields. This section summarizes the principal notational conventions.

Definition 4.1 (Indicator function). For any subset $A \subseteq \Omega$ of a universal set Ω , the **indicator function** of A is the function $\mathbb{1}_A : \Omega \rightarrow \{0, 1\}$ defined by:

$$\mathbb{1}_A(x) \stackrel{\text{def}}{=} \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

More generally, for any logical proposition or predicate P , the **indicator** of P takes the value 1 when P is true and 0 when P is false:

$$\mathbb{1}(P) \stackrel{\text{def}}{=} \begin{cases} 1, & \text{if } P \text{ is true} \\ 0, & \text{if } P \text{ is false} \end{cases}$$

Exm

Example 4.1 (Evaluating set and predicate indicators). Consider the real line $\Omega = \mathbb{R}$, the set of nonnegative numbers $A = [0, \infty)$, and a continuous random variable Y .

1. **Set indicator** $\mathbb{1}_A(x)$:
 - For $x = 3.5$: since $3.5 \in [0, \infty)$, $\mathbb{1}_A(3.5) = 1$.
 - For $x = -2.1$: since $-2.1 \notin [0, \infty)$, $\mathbb{1}_A(-2.1) = 0$.
2. **Predicate indicator** $\mathbb{1}(Y > 5)$:
 - If an observation yields $Y = 7.2$, the predicate $7.2 > 5$ is true, so $\mathbb{1}(7.2 > 5) = 1$.
 - If an observation yields $Y = 4.1$, the predicate $4.1 > 5$ is false, so $\mathbb{1}(4.1 > 5) = 0$.

4.1 Two primary notational paradigms: set vs. predicate notation

The vast majority of indicator notations belong to one of two families: **set notation** or **predicate notation**.

4.1.1 Set notation

In **set notation**, the indicator is tied to a set A , which appears as a subscript:

- $\mathbf{1}_A(x)$ (bold numeral one)
- $\mathbb{1}_A(x)$ (blackboard bold numeral one)
- $I_A(x)$ (capital letter I)
- $\mathbb{I}_A(x)$ (blackboard bold letter I)
- $\chi_A(x)$ (Greek letter chi, historically termed the *characteristic function*)

⁵[proof-writing.qmd](#)

When the function is viewed as a mathematical object in its own right (for instance, as an element of an L^p function space), authors often omit the argument x , writing simply $\mathbf{1}_A$, $\mathbb{1}_A$, or I_A .

4.1.2 Predicate notation

In **predicate notation**, the indicator takes a logical condition, relation, or proposition P directly as its argument or subscript:

- $\mathbb{I}(x \in A)$ or $\mathbb{I}(P)$
- $\mathbf{1}(x \in A)$ or $\mathbf{1}(P)$
- $\mathbb{1}(x \in A)$ or $\mathbb{1}(P)$
- $\mathbb{1}\{x \in A\}$ or $\mathbb{1}\{P\}$
- $I(x \in A)$ or $I(P)$

Predicate notation is especially common in applied statistics and survival analysis, where indicators frequently depend on inequalities involving random variables, such as $\mathbb{1}(T_i \leq t)$ (an event occurring before time t) or $\mathbb{1}(Y_i = 1)$ (a binary outcome).

4.1.3 Equivalence between paradigms

The two paradigms are connected by evaluating the predicate indicator at the membership statement $x \in A$:

$$\mathbf{1}_A(x) = \mathbb{I}(x \in A)$$

Set notation is more natural when the underlying set A has a standard name (such as the support of a distribution or a geometric region). Predicate notation is more natural when the condition involves compound inequalities, such as $\mathbb{1}(0 \leq t \leq u)$.

4.2 Iverson bracket notation

In 1962, Kenneth Iverson introduced a compact notation in the programming language APL, later popularized in mathematics and computer science by Donald Knuth: the Iverson bracket⁶.

The Iverson bracket encloses any mathematical predicate P inside square brackets:

$$[P] \stackrel{\text{def}}{=} \begin{cases} 1, & \text{if } P \text{ is true} \\ 0, & \text{if } P \text{ is false} \end{cases}$$

Under this notation, set membership is written $[x \in A]$, and the Kronecker delta is simply $\delta_{ij} = [i = j]$.

4.2.1 Strengths and limitations of the Iverson bracket

The primary advantage of the Iverson bracket is algebraic conciseness: it converts domain restrictions in sums and integrals into unrestricted operations. For example:

$$\sum_{x \in A} f(x) = \sum_x f(x)[x \in A]$$

However, in statistics and epidemiology, square brackets are already heavily overloaded: they denote closed intervals $[a, b]$, conditional expectations $E[Y \mid X]$, and matrix delimiters. To prevent visual confusion with expectation brackets or intervals, statistical literature predominantly uses $\mathbb{1}$ or I rather than the bare Iverson bracket.

4.3 Summary of indicator notations

Table 3 compares the major notations encountered across the literature.

⁶https://en.wikipedia.org/wiki/Iverson_bracket

Table 3: Notations for indicator functions across mathematical and statistical literature

Notation style	Typical syntax	Primary fields	Notes and potential ambiguities
Blackboard bold 1	$\mathbb{1}_A(x), \mathbb{1}(P)$	Modern probability, mathematical statistics	Unambiguous; distinct from matrices and scalars; standard in this book.
Bold numeral 1	$\mathbf{1}_A(x), \mathbf{1}(P)$	Probability theory, measure theory	Can be confused with a vector of ones $\mathbf{1} = (1, \dots, 1)^\top$.
Blackboard bold I	$\mathbb{I}(x \in A), \mathbb{I}(P)$	Econometrics, machine learning, statistics	Clear predicate notation; avoids confusion with numerals.
Letter I	$I_A(x), I(P)$	Classical statistics, epidemiology	Can be confused with the identity matrix I or Fisher information $\mathcal{I}$.
Iverson bracket	$[P], [x \in A]$	Computer science, discrete mathematics	Very compact, but square brackets collide with intervals and expectation brackets.
Greek letter χ	$\chi_A(x)$	Real analysis, measure theory	Often termed “characteristic function”; collides with the Fourier transform in probability.

4.4 Conventions in this book

In these notes, we standardize on blackboard bold $\mathbb{1}$ via the macros defined in `latex-macros/macros.qmd`:

- `\indic{A}` produces $\mathbb{1}_A$ (set subscript)
- `\indicp{P}` produces $\mathbb{1}(P)$ (predicate in parentheses)
- `\indiccb{P}` produces $\mathbb{1}\{P\}$ (predicate in curly braces)
- `\1{P}` produces $\mathbf{1}_P$ (text numeral with subscript, used in legacy formulas)

Blackboard bold $\mathbb{1}$ is preferred because it avoids all common collisions: it is visually distinct from the scalar 1, the identity matrix I , and the information matrices $(I, \mathcal{I})$.

4.5 Key algebraic properties

Indicator functions translate logical operations on events into ordinary arithmetic on real numbers:

- **Intersection (“and”)**: $\mathbb{1}(A \cap B) = \mathbb{1}(A) \cdot \mathbb{1}(B)$
- **Union (“or”)**: $\mathbb{1}(A \cup B) = \mathbb{1}(A) + \mathbb{1}(B) - \mathbb{1}(A) \cdot \mathbb{1}(B)$
- **Complement (“not”)**: $\mathbb{1}(\neg A) = 1 - \mathbb{1}(A)$
- **Idempotence**: $(\mathbb{1}(A))^2 = \mathbb{1}(A)$
- **Expectation gives probability**: For any event A , the expectation of its indicator is the probability of the event:

$$\mathbb{E}[\mathbb{1}(A)] = 0 \cdot \Pr(\neg A) + 1 \cdot \Pr(A) = \Pr(A)$$

This fundamental identity connects probability theory directly to linear expectation. It provides the mathematical foundation for empirical proportions, survival curve estimators, and regression models for binary outcomes.

5 Stochastic vs. probabilistic vs. random

The terms “stochastic”, “probabilistic”, and “random” are frequently used in statistics and probability theory, often interchangeably in everyday conversation, but they carry nuanced technical distinctions.

5.1 Key distinction: modeling approach vs. phenomena

As noted in Wikipedia⁷:

Stochasticity and *randomness* are technically distinct concepts: the former refers to a modeling approach, while the latter describes phenomena; in everyday conversation these terms are often used interchangeably.

5.2 Definitions

Random describes something that occurs by chance, without a deterministic pattern. It is the most general term, used to describe variables or occurrences whose outcome cannot be predicted precisely, only probabilistically. For example, we speak of “random variables” and “random events”.

Note

The term “random” is sometimes used as shorthand for a uniform distribution (especially the discrete uniform distribution), but it can refer to any probability distribution.

Stochastic comes from the Greek “*στόχος*” (*stókhos*), meaning “aim” or “guess” (see etymology⁸). In mathematics, a **stochastic process** is formally defined as a collection of random variables indexed by time or space. The term is almost always used in the context of processes or systems evolving in time or space under uncertain rules. Note that in probability theory, “stochastic process” and “random process” are synonyms (Adler and Taylor 2009; Stirzaker 2005; Kallenberg 2002).

Probabilistic refers to any model, reasoning, or method that explicitly involves probability theory. Probabilistic models assign probabilities to events or outcomes; they focus on quantifying and reasoning about uncertainty based on known or estimated distributions. While all stochastic models are probabilistic (since they use probabilities), not all probabilistic models need to describe processes evolving in time.

5.3 Summary of usage

Table 4: Comparison of “random”, “stochastic”, and “probabilistic”

Term	What it describes	Typical use	Example
Random	Single variable or event	Random variable, random outcome	Coin toss, die roll
Stochastic	System or process in time/space	Stochastic process	Stock price evolution, Markov chain
Probabilistic	Approach/model using probability	Probabilistic model/reasoning	Bayesian inference, regression

While some sources treat “stochastic” and “random” as practically synonymous, the academic preference is to use “random” for variables and events, and “stochastic” for processes, especially to highlight temporal or spatial structure in the modeling.

5.4 Additional resources

- Wikipedia: Stochastic⁹

⁷<https://en.wikipedia.org/wiki/Stochastic>

⁸<https://www.etymonline.com/search?q=stochastic>

⁹<https://en.wikipedia.org/wiki/Stochastic>

- Wikipedia: Stochastic process¹⁰
- Mathematics Stack Exchange: What's the difference between stochastic and random?¹¹
- Cross Validated: Probability model vs statistical model vs stochastic model¹²

6 Why is notation in probability and statistics so inconsistent and disorganized?

In grad school, we are asked to learn from increasingly disorganized materials and lectures. Not coincidentally, as the amount of organization decreases, the amount of complexity increases, the amount of difficulty increases, the number of reliable references decreases, and the amount of inconsistency in notation and content increases (both between multiple references and within single references!). In other words, as you approach the cutting-edge of most fields, you start to encounter into content that hasn't been fully thought through or standardized. This lack of clarity is unfortunate and undesirable, but it is understandable and inevitable.

It's worth noting that calculus was formalized in the 1600s¹³, elementary algebra was formalized around 820¹⁴, and arithmetic even earlier¹⁵. And calculus still has several competing notation systems¹⁶. In contrast, the field of statistics only emerged in the late 1800s and early 1900s¹⁷, so it's not surprising that the notation and terminology is still developing. Generalized linear models were only formalized in 1972 (Nelder and Wedderburn (1972)), which is very recent in terms of the pace of scientific development¹⁸.

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¹⁰https://en.wikipedia.org/wiki/Stochastic_process

¹¹<https://math.stackexchange.com/questions/114373/whats-the-difference-between-stochastic-and-random>

¹²<https://stats.stackexchange.com/questions/421462/probability-model-vs-statistical-model-vs-stochastic-model>

¹³https://en.wikipedia.org/wiki/Leibniz%27s_notation

¹⁴<https://en.wikipedia.org/wiki/Al-Jabr>

¹⁵<https://en.wikipedia.org/wiki/Arithmetic#History>

¹⁶https://en.wikipedia.org/wiki/Notation_for_differentiation

¹⁷https://en.wikipedia.org/wiki/History_of_statistics#Development_of_modern_statistics

¹⁸https://en.wikipedia.org/wiki/The_Structure_of_Scientific_Revolutions